

Design a Solution for PASS FIRST TIME Driving School

Solution and Diagram

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1 Introduction

In this report, a detailed design of the management system of PASS First Time Driving School is provided. The key objective of such a system is to substitute the existing manual process where hand written record cards, paper documents, and disjointed tracking systems are heavily used. Not only these obsolete methods slow day to day processes they also elevate the possibilities of mistakes, lost documents and lacks of regular communications between the pupils and the teachers(Sommerville, 2023).

The suggested computerised system will modernise the whole procedure, and such a system presents effective digital solutions to the process with lesson booking, cancellations, monitoring payments, and scheduling inventors. It will also have an organised record keeping system where all the critical data regarding the students and instructors will be stored in an organised and safe way(Schwalbe, 2021).

Besides that, the design focuses on clean and usable interface to make both administrative staff and instructor to use the system with bare minimum training. The solution will also seek to provide the major management insights using various reporting tools to help the driving school to maintain performance, student progress, lesson review statistics and better management of resources. In general, the system will be aimed at streamlining the operations, minimizing the manpower and facilitating smarter decision-making in the organisation(Petersen, Vinter Gencel, 2022).

2 Part A: Detailed Design Documentation

2.1 Software Development Life Cycle (SDLC) Models Assessment

It is important to the selection of an appropriate Software Development Life Cycle (SDLC) model the success of the new system. In the PASS first time driving school project, two models that we will consider are the Waterfall Model and the Agile Model(Kotonya Sommerville, 2022).

2.1.1 The Waterfall Model

The waterfall model is a conventional, serial software development model where each step has to be completed, after which the process moves on to the subsequent step. This flow is linear and this is what makes the entire process predictable and orderly(Gotel, Morris Dahan, 2020). Due to its organization, the model has been viewed as simple to plan, control and administer. It is most appropriate in small projects

or the ones with a clear objective and requirements that are not volatile. This model may provide an easy way to conceive to deliver when the scope is evident on the very outset and the anticipated result remains relatively constant over time.

Benefits Much of the transparency and high sense of organization are one of the main strengths of the Waterfall Model. Each phase has a distinct order and this enables teams to avoid confusion on progress. The rigorous reports that are generated in every stage, including requirements list, design charts, test procedures, and user documentation, formulate an all encompassing project log(Whitten Bentley, 2020). This documentation will become very handy whenever new team members are added, as well as in cases where work is to be passed on or even when maintenance of the system is required in future. It can also give concrete justification to the decisions made on designing, which helps simplify the whole development process to audit or revise.

Disadvantages The model however has some major drawbacks. The most evident disadvantage is that it is not flexible. The requirements are supposed to be completely defined in the initiation of the project and this leaves the project with minimal room to change later on. In case it requires any change in the course of development, it might be expensive and time consuming since they might require to retrace and rework previous stages(Hoffer, George Valacich, 2023). Another aspect of concern is that it could be that the end product would not suit the real needs of the users particularly when the needs of the users change in the long development process. As user feedback is normally gathered at the final stages, there is a probability of providing a solution which does not match expectation by the time the solution is ready.

Waterfall Model Suited to the Driving School System The Waterfall Model is moderately suitable. The driving school system requirements including booking cancellations, and books, are comparatively clear and constant. However, the system's size small to medium and a dogmatic strategy may slow the process of delivering a working system(Shneiderman, Plaisant Cohen, 2020).

2.1.2 The Agile Model

Agile Model is a software development model that follows the iteration and incremental delivery. Agile does not take the work through a long, stiff structure since they finish project through a sequence of very small, manageable cycles known as sprints. Sprints deliver small yet working amounts of the software regularly, which enables the team to discuss the progress regularly and redefine their priorities when required(Krug,

2022). This approach puts a high value on flexibility, highly working in concert with customers, and improvement in delighting customers. The model fits well within dynamic environments where change is anticipated and it has to be accommodated in the most fewest time possible as requirements can be integrated at any point.

Strengths Agile has several significant strengths and flexibility is one of them. Today software projects are frequently confronted with change of clients expectations, technological change, and random market needs and Agile is tailored to managing such changes effortlessly (Silberschatz, Korth Sudarshan, 2023). The frequent feedbacks made by the client will keep the system under development in track with what the client actually needs as opposed to assumptions made at the onset. Continued communication minimizes the chances of getting confused and ensures that the project is directly dependent on actual user value. The other advantage is the ability to deliver working features early and often; even during an initial stage, the stakeholders can view a practical output, which will build confidence and allow the business to be of value without having to wait to the final release.

Drawbacks Agile also has its disadvantages despite its advantages. This may become a problem as there is no large amount of formal documentation and in cases that the system is supposed to be taken care of in a long term or when new engineers come into the picture and require historical data (Jansen, Schots Van de Walle, 2020). Since Agile is a team-based methodology, which involves regular communication, it requires regular participation by the clients and other stakeholders. Such a commitment can be hard to sustain during the sustained projects of development and may be a load to those teams that are limited in the available resources. Moreover, the lack of discipline in the project will lead to losing track as new changes will frequently occur, or project priorities will be uncertain.

Fits in Driving School System The Agile method is very appropriate in the case of a Driving School Management System. There is usually a need in such a system to develop over time and have functionality such as scheduling of lessons, tracking of student progress, payment system, instructor availability and new features that could be proposed by the users (Pressman Maxim, 2021). Agile being iterative enables the school to start with simple functionalities and as more is obtained the functionalities are developed according to the actual feedback. The brief development periods allow having chances to check the usability, work on the interface, and transform the working process to the way that instructors, students, and administrators actually work. This

makes sure the system is constantly refined and in accordance to the practical needs of the school as they expand.

2.2 Gathering of Information: interviews and Surveys

In order to design a perfect and efficient system, we need to obtain an in-depth information first of the people that will use it. Interviews and surveys will mainly be used during this process.

2.2.1 Interviews of the Stakeholders

Interviews become useful in terms of understanding the business needs and objectives. By we can speak directly to the driving school owner, office personnel and instructors discover stepwise processes, areas of pain, and necessary needs that could not be from documents it is obvious(Larman Basili, 2020). As an illustration, the process can be known through an interview to manage an instance of the cancellation of a lesson, who should be informed, and what documentation must be updated. It is this face-to-face interaction that contributes to the development of a system which actually assists the business's objectives.

2.2.2 Surveys for User Feedback

One of the best methods to have a larger number of people give feedback is through the use of survey whereby in this case is the student drivers(Kroenke Boyle, 2022). Although students are not the major consumers of the management system, their feedback on reservation choices, communication channels is useful. A survey can easily either find out popular frustrations in the present booking process or propose functions that would enhance the experience which automatically makes them have a good system(Schwalbe, 2021).

2.3 Data documentation: Data Flow Diagrams and Flowchart

After the requirements gathering, the logic and data flow in the system needs to be followed clearly documented.

2.3.1 Data Flow Diagrams (DFD)

Flow charts indicate the flow of information within the system.

Context Diagram Context Diagram The Context Diagram is the overall picture of the PASS FIRST TIME Driving School System, which demonstrates the system in

relation to other users (pupils who make bookings and payments, instructors who manage lessons, management which tracks the operations) (Petersen, Vinter and Gencel, 2022; Gotel, Morris and Dahan, 2020). It shows the flow of information in its movement both between the system and these parties, as well as their roles and system boundaries.

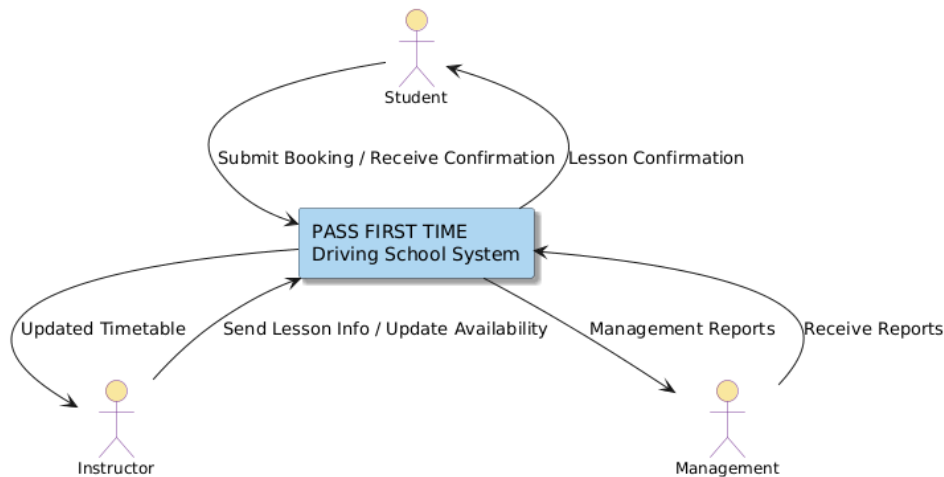


Figure 1: DSS context Demo Diagram of the Driving School System

Level 1 Diagram Level 1 Diagram divides the PASS FIRST TIME Driving School System into essential function processes, bookings, student records, and reports (Whitten and Bentley, 2020). It shows the flow of information regarding these processes and how they interact with data stores, which explain how the system is built to achieve the requirements of students, instructors and management (Hoffer, George and Valacich, 2023).

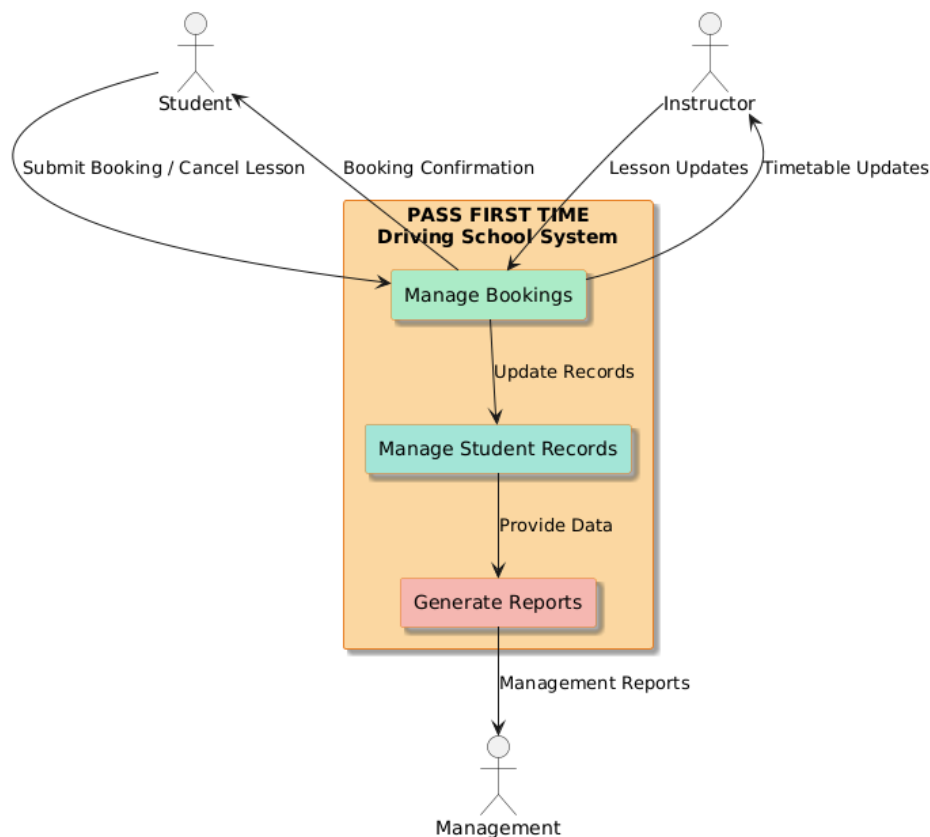


Figure 2: Demo Level 1 Data Flow Diagram

Level 2 Diagram (Example: Manage Bookings) The Level 2 Diagram describes a particular process within the PASS FIRST TIME Driving School System, and it splits bookings into smaller sub-processes such as lesson booking, cancellations, and schedule changes (Shneiderman, Plaisant and Cohen, 2020; Krug, 2022). It can be used to demonstrate the flow of data between these activities and the related data stores to understand the inner-workings of the booking process.

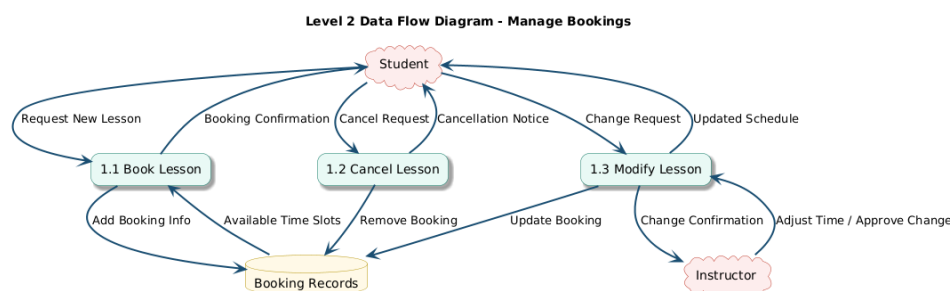


Figure 3: Level 2 Demo Data flow Diagram (Manage Bookings)

2.3.2 Flowchart

A flowchart is an illustrative representation of the processes and choices of a particular procedure, e.g. the procedure in order to book a new lesson.

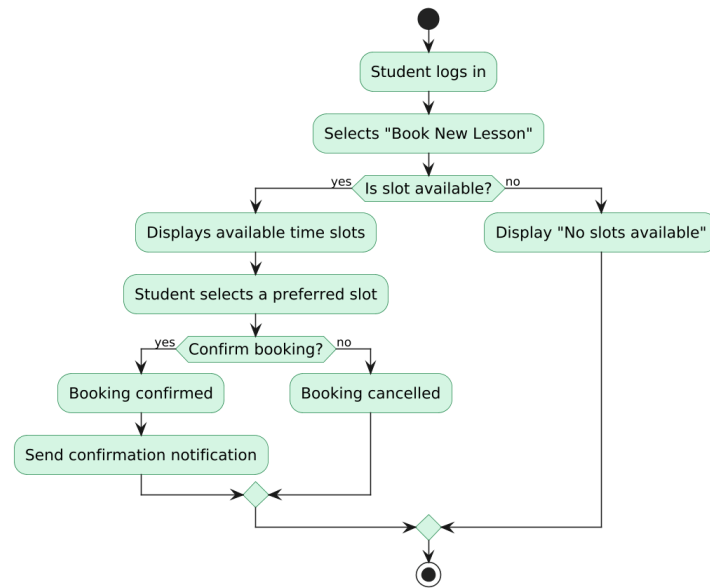


Figure 4: Demo Flowchart of new lesson booking process

2.4 User Interface and Hardware/software Requirements

The user interface (UI) should be user friendly and effective to the office personnel and instructors.

2.4.1 Data Entry Forms and Layout

New Student Registration data will be the primary data entry form. The layout will be clean, with a standard three-column design to squeeze in as many columns as can be (Jansen, Schots Van de Walle, 2020).

NEW STUDENT REGISTRATION FORM

Personal Information

First Name *
Enter first name

Last Name *
Enter last name

Date of Birth *
DD/MM/YYYY

Email Address *
email@example.com

Contact Number *
Enter phone number

Emergency Contact
Enter emergency contact

Course Details

Allocated Instructor *
Select instructor

Course Type *
Select course type

CLEAR SAVE CANCEL

Figure 5: Demo Integration of New Student Registration Form

The proposed Fields Important student information will be saved in the system like Name, Contact Number, Email, Date of Birth, and Assigned Instructor to maintain the learner profiles effectively (Garbuno and Sanchez, 2023). Data entry will be done with convenient interfaces: text boxes with simple information, a date picker to enter the Date of Birth and a drop-down with choices of the Instructor. This guarantees precision, uniformity, and dependable enrollment of students (Larman and Basili, 2020).

2.4.2 Hardware and Software Requirements

The system must have the capability of operating on the normal and modern office equipment to minimize costs.

Hardware

- **Server:** A dedicated small server or a virtual machine based in the cloud to host the database and application.
- **Client Devices:** Office staff will use standard desktop computers or laptops which might be instructor tablets(Krug, 2022).

Software

- **OS:** Windows 10/11 or a Linux is a stable and secure OS distribution.
- **Database:** A credible relational database management system (RDBMS) like PostgreSQL or MySQL(Whitten Bentley, 2020).
- **Application:** An application created as a web-based solution (e.g., Easy access and maintenance Python/Django or Node.js/Express).

3 Part B: System Evaluation and Improvement

3.1 System Success and Failure Factors

Success of the new PASS FIRST TIME Driving School system relies upon a number of determining factors, starting with project management up to user adoption.

3.1.1 Risk, Budget and Time constraints

Effective project planning is crucial to avoid risks like scope creep, delays, and budget overruns. Agile methodology mitigates these issues by breaking work into small increments, allowing frequent review and adaptation. This approach improves control, transparency, and alignment with business goals while reducing the likelihood of delays and mismanagement(Garbuno Sanchez, 2023)..

3.1.2 Costs, intrapreneurial time, and organisational policies

The high-upfront price of the system can be explained by the long-term advantages, such as less manual labor, fewer mistakes, and more expedient business mechanisms that result in more quality decision-making (Kotonya and Sommerville, 2022). The development has to be improved on time and without deteriorating the quality. Adherence to the law on data privacy and security brings a legal guarantee and fosters the trust of the users (Pressman and Maxim, 2021).

3.1.3 Organizational Skills and System Regeneration

The effectiveness of the system will depend on how the staff will utilize it. Comprehensive training is essential. In addition, the system should be user-friendly. If the school's current IT personnel do not have the talent to deal with the new technology, they will have to be supported externally, which increases the overall cost(Larman Basili, 2020).

3.1.4 Management and User Involvement of Change

System failure is usually due to resistance by the user which in most cases is due to fear of new computerized processes. Concise explanation of the advantages of the system, including efficiency, reduction of errors, and saving of time, will facilitate a feeling of ease (Sommerville, 2023). Involving end-users in design and development makes the system relevant to the end-users, creates a sense of ownership and promotes acceptance and cooperation, as well as adoption in the long term (Whitten and Bentley, 2020).

3.2 Design Assessment to User Requirements

The identified core user needs in the suggested design are met directly initial investigation.

3.2.1 Meeting User Requirements

- **Booking and Cancellation:** Manage Bookings process offers the following through the system an easy and quick means of booking and cancelling lessons, as opposed to the slow and problematic phone and paper-based method.
- **Record Keeping:** Maintaining all student and instructors information at a central database can guarantee speedy access and convenience in editing, removing any physical records cards (Schwalbe, 2021).
- **Reporting:** The system will provide complete reporting menus, comprising of The weekly schedules which serve to meet with the need of the management directly operational oversight.

3.2.2 Recommendations on the potential improvements and further developments

At some point later on, the system can be improved to bring even more value:

- **Online Student Portal:** This will enable students to reserve their lessons to be handled online would greatly contribute to the workload of the office personnel.
- **Integration of payments:** introduction of a lesson payment processing feature via the system would simplify the financial procedure (Gotel, Morris Dahan, 2020).

- **Mobile App Instructor:** an instructor specific mobile application to update would promote efficiency in the lesson progress and would present a schedule in real-time the field.

4 Conclusion

The proposed computerized system of PASS FIRST TIME Driving school is target to offer the school with a modernized, efficient and all round solution to the existing operational obstacles of the school. The system will simplify and automate the core operations of the company without compromising on accuracy and good uniformity by eliminating the existing manual work, which has been very cumbersome as the data is stored on physical record cards, paper sheets and requires a lot of time in conducting the manual work(Whitten Bentley, 2020). The system is designed based on fast and flexible approach as a result of which it is capable of adapting to new requirements and being able to accommodate future improvements without significant disruption.

One of the notable characteristics of the system is that it is user friendly, and this highlights the fact that it is easy to use among all forms of users, such as office staff, instructors, and the management staff. This guarantees that those who will be communicating with the system can efficiently work with the system quickly, naturally and making minimum errors(Hoffer, George Valacich, 2023). In addition to the automation of the simple processes like booking of students, cancellation of lessons, assignment of instructors and record keeping, the system also incorporates multi-faceted management information systems. These reporting and analytics features enable the administration at the school to track the operations in real time, make sound decisions, and find the areas of improvement. Combining all these features, the system is not only streamlining the everyday operations but enhancing the capacity of the school in how it manages its resources, reaches a higher customer satisfaction, and results in the long-term expansion and operational productivity(Schwalbe, 2021).

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